

SIMATS ENGINEERING

ASSESSMENT TOOL

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

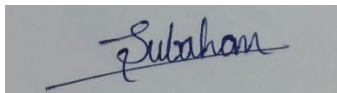
Annexure A

Dry Run Evaluation

Code of Conduct:

I T. Subahan(192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Subahan' with a stylized flourish at the end.

Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/ \ / \

D E F G

Task: Starting from node A, perform a Breadth-First Search (BFS). Complete the following table.

Iteration	Queue	Visited Node
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1		
---	--	--

2		
---	--	--

3		
---	--	--

4		
---	--	--

5		
---	--	--

6		
---	--	--

7		
---	--	--

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

A

/ \

B C

/ \ / \

D E F G

BFS visits nodes level by level from left to right.

1. BFS Table

Iteration	Queue before processing	Visited Node	Queue after processing
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

2. BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

3.Queue Contents After Each Iteration

Initial Queue: [A]

After Iteration 1: [B, C]

After Iteration 2: [C, D, E]

After Iteration 3: [D, E, F, G]

After Iteration 4: [E, F, G]

After Iteration 5: [F, G]

After Iteration 6: [G]

2. Depth-First Search (DFS) Using the same graph,

```
A
 /\
B C
 /\ /\
D E F G
```

**Perform Depth-First Search (DFS) starting from A.
Complete the table.**

```
A
 /\
B C
 /\ /\
D E F G
```

Iteration	Stack	Visited Node	Stack After Processing
1	A	A	B, C
2	B, C	B	D, E, C
3	D, E, C	D	E, C
4	E, C	E	C
5	C	C	F, G
6	F, G	F	G
7	G	G	Empty

DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Stack Contents After Each Iteration

Initial Stack: [A]

After Iteration 1: [B, C]

After Iteration 2: [D, E, C]

After Iteration 3: [E, C]

After Iteration 4: [C]

After Iteration 5: [F, G]

After Iteration 6: [G]

After Iteration 7: [Empty]

3. Initial State :

1 3 6

5 0 2

4 7 8

Goal State :

1 2 3

4 5 6

7 8 0

Task

Trace Breadth-First Search (BFS) until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1

2

3

4

5

Questions

- 1. Which node is expanded in each iteration?**
- 2. How many states are generated in total?**
- 3. Write the complete solution path from the initial state to the goal state.**
- 4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?**

Given

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

3. BFS Table — First 5 Iterations

Iteration	Current State Expanded	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	[S0]	[S1, S2, S3, S4]
2	1 0 6 / 5 3 2 / 4 7 8	[S1,S2,S3,S4]	[S2,S3,S4,S5,S6]
3	1 3 6 / 5 7 2 / 4 0 8	[S2,S3,S4,S5, S6]	[S3,S4,S5,S6,S7,S8]
4	1 3 6 / 0 5 2 / 4 7 8	[S3,S4,S5,S6, S7,S8]	[S4,S5,S6,S7,S8,S9,S10]
5	1 3 6 / 5 2 0 / 4 7 8	[S4,S5,S6,S7, S8,S9,S10]	[S5,S6,S7,S8,S9,S10,S11,S 12]

4. Node Expanded in Each Iteration

Iteration 1 → Initial state

Iteration 2 → 1 0 6 / 5 3 2 / 4 7 8

Iteration 3 → 1 3 6 / 5 7 2 / 4 0 8

Iteration 4 → 1 3 6 / 0 5 2 / 4 7 8

Iteration 5 → 1 3 6 / 5 2 0 / 4 7 8

5. Total Number of States Generated

Using BFS with duplicate states avoided:

500 unique states are generated/stored before the goal is reached.

The goal is reached when it is expanded at the 310th expansion.

6. Complete Shortest Solution Path

The shortest path contains 8 moves:

Initial

1 3 6

5 0 2

4 7 8

↓ Right

1 3 6

5 2 0

4 7 8

↓ Up

1 3 0

5 2 6

4 7 8

↓ Left

1 0 3

5 2 6

4 7 8

↓ Down

1 2 3

5 0 6

4 7 8

↓ Left

1 2 3

0 5 6

4 7 8

↓ Down

1 2 3

4 5 6

0 7 8

↓ Right

1 2 3

4 5 6

7 0 8

↓ Right

1 2 3

4 5 6

7 8 0

Right → Up → Left → Down → Left → Down → Right → Right

Number of moves = 8

7. Why Does BFS Find the Shortest Solution?

BFS explores the state space level by level:

Level 0 → Initial state

Level 1 → States 1 move away

Level 2 → States 2 moves away

Level 3 → States 3 moves away

...

Hence, BFS always finds the shortest solution in an unweighted 8-puzzle problem.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.